Dr. Tatyana Sitnova

Personal Details

Position: Researcher, The Division of Stellar Spectroscopy, the Institute of Astronomy of the Russian Academy of Sciences (INASAN), Pyatnitskaya 48, Moscow, RUSSIA, 119017
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Publications/citations: 65/1206
Date of birth: January 27, 1990
Nationality: Russian
Country of residence: the U.S. lawful permanent resident

Research Areas

- High-resolution stellar spectroscopy
- Non-LTE spectral line formation
- Chemical evolution, neutron-capture nucleosynthesis
- Galactic archaeology, stellar populations
- Chemically peculiar metal-poor stars

Aims

- Developing non-LTE techniques for accurate stellar spectroscopy studies
- Identifying the origins of chemical elements via non-LTE abundances and isotope ratios
- Getting observational data for inderstanding nucleosynthesis and Galactic chemical evolution

Education

2006 - 2012	Student at Moscow State University, Faculty of Physics, Department of Astronomy
February 2012	Diploma in Astrophysics, Sternberg Astronomical Institute, Moscow State University (SAI MSU) Title: Non-LTE line formation for O I and Ca I-II in A-K spectral type stars Supervisor: Dr. L.I. Mashonkina
2012 - 2016	Postgraduate student at SAI MSU
February 2017	Ph.D. in Astrophysics, SAI MSU Title: Determination of atmospheric parameters and chemical element abundances of B-K spectral type stars based on the non-LTE line formation PhD advisor: Dr. L.I. Mashonkina

Employment

2016 - present	Researcher/Senior Researcher, INASAN
2012 - 2016	Junior Researcher, INASAN Laboratory Assistant, SAI MSU

Selected Funded Projects and Awards

2020 – 2022 Non-LTE abundances of Zn and Fe and the Galactic [Zn/Fe] trend; Theoretical Physics and Mathematics Advancement Foundation 'BASIS' (an individual grant)
2018 – 2019 Chemical evolution of dwarf galaxies, the Milky Way satellites; Russian Academy of Science foundation
2015 – 2017 Spectroscopy of BAF-type stars; Russian Foundation for Basic Research
2015 – 2017 Development of collision theory and its applications for calculations of atomic and molecular data on inelastic collision processes; Russian Scientific Foundation
2013 – 2016 Systematic NLTE study of the -2.6 < [Fe/H] < 0.2 F and G dwarfs in the solar neighbourhood; Russian Foundation for Basic Research

- 'The best studies of young researchers, 2017' prize, the Department of Physics of Russian Academy of Sciences
- A poster prize (Lecturers choice), XXIX Canary Islands winter school 'Applications of radiative transfer to stellar and planetary atmospheres', November 2017

Selected Talks

Over 60 talks at conferences, meetings, and workshops. Ten selected talks are listed.

- "Tracing s-, i-, and r-processes through barium isotope ratios in VMP stars", The R-Process Alliance workshop, 4-7 November 2025, Livermore
- "Barium isotope ratio in very metal-poor stars as a key to a puzzle of light neutron-capture element synthesis at the earliest epoch", s, i & r Element Nucleosynthesis (sirEN) Conference, Giulianova, June 8-13, 2025
- "Galactic Archaeology via high-resolution spectroscopy of stellar streams", San Francisco State University colloquia, October 9, 2023
- "Chemical composition of metal-poor stars in Coma Berenices and Segue 1 systems as a key to nucleosynthesis in the first stars" at the EWASS, Special Session SS9: Dwarf galaxies near and far: bridging numerical simulations with observations, Lyon, June 24-28 2019
- "Influence of inelastic collisions with hydrogen atoms on NLTE abundance determination", invited talk at the international conference Stars, planets, and their magnetic fields, Saint-Petersburg, September 17-21, 2018
- "Atmospheric parameters and magnesium and calcium NLTE abundances for a sample of 16 ultra metal-poor stars" at the AAS Meeting #232 Stellar abundances in dwarf galaxies, Denver, June 2018
- "NLTE abundances of C, O, Mg, Ca, Ti, and Fe in BAF-type stars" at the IAU General Assembly, division G Stellar Physics, Vienna, August 20-31 2018
- "Determination of atmospheric parameters for the nearby -2.6 < [Fe/H] < 0.2 F-G dwarfs", National Astronomical Observatories colloquia, Beijing, October 8, 2014
- "Influence of Departures from LTE on Oxygen Abundance Determination in the Atmospheres of A–K stars" at the IAUS 298 "Setting the scene for Gaia and LAMOST", Lijiang, May 20-24, 2013
- "The r- and s-process contributions to heavy element abundances of a halo star HD 29907" at the SCOPES Workshop "Heavy elements in galactic chemical evolution and NLTE effects", Odessa, August 31 - September 1, 2011

Scientific Collaborations

- The High-Resolution spectroscopic program on the Galactic Origins of elements (HR-GO, PI: Z.Yuan)
- Chemical Evolution of R-process Elements in Stars (CERES, PI: C. Hansen); outside contributor
- The Pristine survey (PI: E. Starkenburg, N. Martin); outside contributor
- The formation of the Milky Way halo and its dwarf satellites; a NLTE-1D abundance analysis (PI: L. Mashonkina)
- Chemical elements as tracers of the evolution of the cosmos (ChETEC, the European cooperation in science and technology program, PI: R. Hirschi), COST, 2017-2020
- The formation and Evolution of the Galactic Halo, International Team supported by The International Space Science Institute (ISSI, Switzerland, PI: D. Romano), 2016-2017
- Stars, Stellar Explosions and the Origin of the Elements (SCOPES, PI: F.-K. Thielemann), 2014-2017
- First stars in dwarf galaxies, (ISSI, Switzerland, PI: P. Jablonka), 2012-2014
- Synthesis of heavy elements in core collapse supernovae and their imprint on galactic chemical evolution (SCOPES, PI: F.K. Thielemann), 2011-2013

Teaching and Science Popularization

2022 - 2025 MS/BC advisor for Ekaterina Niagina at SAI MSU
2025 Diploma Thesis of E. Niagina got an award from "Frontiers of Science" foundation[1]
2023 Summer internship for Ekaterina Ageeva
2018 - 2021 MS/BC co-advisor for Maria Neretina (advisor Dr. L.I. Mashonkina)

2014 - 2015 Lab Tutor in Astrophysics at Moscow State University
2011 - 2016 Guide at Moscow planetarium

- "Early chemical evolution of the Galaxy and its dwarf satellites", lecture for the planetarium staff school, Kaluga, March 1-4, 2022
- "Galactic archaeology", public talk at Moscow planetarium, October 12, 2022
- "The basics of stellar and exoplanetary spectroscopy and photometry", lecture for the summer space school, August 16-23, 2020, Moscow
- "The origin of chemical elements", science slam talk, August 17, 2014, Polytechnic Museum, Moscow
- "Stellar spectroscopy", science slam talk, December 13, 2014, Polytechnic Museum, Moscow

Additional Activities

I am a referee for the Astrophysical Journal, Astronomy Letters, etc. since 2018.

I co-observed with Zhen Yuan for high-resolution spectral observations with the MIKE/Magellan spectrograph/telescope in August 2022.

I keep track of non-LTE studies and compile them in an online database:
https://non-lte.com/bibliography.html

[1] Text in Russian:
https://msu.ru/news/novosti-mgu/gildiya-rubezhi-nauki-nagradila-studentov-fizicheskogo-fakulteta-mgu.html